

Multi-view Hierarchical Graph Contrastive Learning based on Asynchronous Asymmetric Structure

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Abstract

Contrastive learning has strong generalization ability and the capability to learn automatically without labeled information. However, it still faces challenges such as insufficient feature diversity, a lack of multi-level semantics, and the balance between tolerance and consistency. To address these challenges, This study propose a Multi-view Hierarchical Graph Contrastive Learning method. First, a new view is generated through a diffusion matrix to provide multi-view data for contrastive learning. Then, these multi-view data are fed into an asynchronous asymmetric network structure, specifically using graph network models to learn diversified features. Next, we adopt a self-designed hierarchical contrastive learning

framework, constructing a three-level contrastive loss for joint optimization of nodes, subgraphs, and global graphs. Meanwhile, we introduce alignment and consistency and appropriately adjust the loss function through a temperature coefficient. Ultimately, the model achieves excellent classification performance on multiple datasets through node classification and graph classification tasks.

CCS Concepts

• **Computing methodologies** → **Machine learning**; **Machine learning algorithms**; **Feature selection**.

Keywords

Multi-view, Hierarchical, Graph Contrastive Learning, Asynchronous Asymmetric Structure.

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1 Introduction

Graph-structured data, prevalent in domains such as social networks, molecular chemistry, and recommendation systems, inherently encapsulate complex relational patterns and hierarchical semantics. To harness such data, graph representation learning has emerged as a pivotal paradigm, with contrastive learning (CL) gaining prominence due to its ability to learn discriminative features without labeled supervision. By maximizing mutual information between augmented views of data, CL demonstrates strong generalization capabilities and robustness against noise. Siamese Networks [9] and Triplet Networks [21] represent early contrastive learning algorithms, introducing complex similarity metrics and deep metric learning models, respectively. In recent years, contrastive learning has seen rapid development, introducing a series of outstanding algorithms such as MSSGCL[30], SimGRACE [49], and Momentum Contrast (MoCov3) [7]. These algorithms enhance efficiency by altering contrastive loss methods, introducing new techniques, or improving network architectures.

Meanwhile, Graph Neural Networks (GNNs) have gained significant attention and application across multiple domains, such as traffic flow prediction [8], object pose estimation [58], and traffic forecasting [47]. GNNs have also evolved in terms of training methods and graph data handling, transitioning from supervised to unsupervised networks capable of processing complex graph structures. Traditional GNNs typically rely on task-specific labeling for robust learning, which presents challenges for processing data with complex labeling requirements, such as language, video, and 3D point cloud data [56]. Additionally, labeling work in downstream tasks and biomedical information domains incurs considerable costs. To address these challenges, graph contrastive learning [13, 44, 53] provides a powerful solution by automatically generating features based on the inherent structure and attributes of the data, enabling training without reliance on labeled data, and applying the learned features to a wide range of downstream tasks.

However, existing graph contrastive learning frameworks face critical limitations: (1) insufficient feature diversity arising from simplistic augmentation strategies, (2) neglect of multi-level semantics (e.g., node, subgraph, and global graph interactions), and (3) the unresolved trade-off between tolerance for local variations and consistency in global representations. These challenges hinder the model's capacity to capture comprehensive structural and semantic information, thereby limiting its applicability to complex real-world scenarios.

To address these gaps, we propose Multi-view Hierarchical Graph Contrastive Learning (MHGCL), a novel self-supervised framework that integrates dynamic multi-view generation, asynchronous asymmetric learning, and hierarchical contrastive optimization. First, we design a dynamic multi-view generation model that leverages semantic-aware graph diffusion to produce diverse views. Unlike static diffusion matrices, our dual-channel diffusion engine combines local semantic preservation (via heat-kernel diffusion and graph attention) with global topology propagation (via centrality-weighted PageRank variants), enabling adaptive and context-aware view generation. Second, we introduce an asynchronous asymmetric architecture that employs two distinct graph network models

and adopts differentiated gradient propagation strategies to decouple local and global scale feature learning. A momentum-based coordination mechanism ensures stable training by balancing real-time gradient updates in the main network with momentum-driven updates in the auxiliary network. Third, we devise a hierarchical contrastive learning framework that jointly optimizes node-, subgraph-, and graph-level semantics through three-tiered contrastive losses. By dynamically adjusting loss weights via a meta-network, our model prioritizes node-level consistency in early training stages and shifts focus to global structure alignment in later phases. Additionally, we incorporate alignment and uniformity metrics to harmonize tolerance and consistency, while temperature coefficients adaptively regulate penalty strengths for hard negative samples. The main contributions are summarized as follows:

Dynamic Multi-view Generation: The dual-channel diffusion engine fuses local-global structural features via combining Heat-kernel diffusion with Graph Attention and integrating the centrality-aware Personalized PageRank variant, overcoming limitations of traditional static augmentation.

Asynchronous Asymmetric Learning: An asynchronous asymmetric structure with two distinct graph network models is proposed, decoupling local-global feature learning via a differentiated gradient propagation strategy and introducing a momentum-based gradient coordination methodology.

Hierarchical Contrastive Optimization: The three-level contrastive loss system integrates node, subgraph, and global graph semantics, ensuring comprehensive feature learning.

2 Related Works

Graph Neural Networks (GNNs) process graph-structured data and excel in graph analysis tasks. Contrastive learning enhances GNNs by enabling self-supervised learning with low annotation costs. We review three aspects: contrastive learning, classical GNNs, and graph contrastive learning (GCL).

2.1 Contrastive Learning

Contrastive methods learn representations via autoregressive modeling, nonlinear transformations, and momentum updates. CPC[36] uses probabilistic contrastive loss, while SimCLR[6] employs learnable nonlinear transformations. SwAV[4] combines clustering and consistency. Asymmetric architectures address collapse: MoCo[20] uses momentum encoders, BYOL[17] avoids negative pairs. For graphs, CGMP-GL[42] integrates generative-contrastive ideas, CGRL[40] combines adversarial reconstruction with information bottlenecks, and TGSL[55] learns temporal graph structures via sequence prediction.

2.2 Classic Graph Networks

Classic GNNs evolved from GCN[27], addressing node classification and link prediction. GCN aggregates neighborhood features via linear transformations. GAT[43] introduces attention to weight neighbor importance. GraphSAGE[18] samples neighbors for scalability. DCNN[3] captures spatial and global relationships via graph convolutions, while GAP[2] adaptively pools nodes to preserve graph structure.

2.3 Graph Contrastive Learning

GCL methods include multi-view, mutual information, and hierarchical approaches. Multi-view methods like MVGRL[19] contrast diffusion and adjacency views, while HDMI[24] extends to heterogeneous graphs. DMGI[37] optimizes multi-relational embeddings via consensus regularization. GRAND[5] uses diffusion PDEs for dynamic semantics.

Mutual information methods include DGI[44] (maximizing local-global mutual information) and GMI[38] (extending mutual information to graphs). GRACE[57] combines mutual information with triplet loss, and InfoGCL[51] reduces task-irrelevant mutual information. PT-HGNN[23] pre-trains via node/pattern-level contrast.

Hierarchical approaches capture multi-scale structures: DIFFPOOL[54] learns differentiable clustering, HiGCN[22] combines attention with hierarchical convolutions, and HGCL[25] contrasts node/graph-level semantics. HSL-RG[26] integrates global and local molecular representations.

3 Method

3.1 Overall Architecture of the Model

Figure 1 illustrates the overall architecture of the model. The architecture consists of the following components: data augmentation, which leverages semantic-aware diffusion to enhance the data, overcoming the limitations of traditional fixed-parameter diffusion matrices and constructing a dynamic multi-view generation model; asynchronous asymmetric structure, including two different GNNs, a shared-weight Multi-Layer Perceptron (MLP), and a graph pooling layer; mutual information, involving three different contrastive methods at different levels for graph feature representation and node feature representation, along with distinct mutual information discriminators.

3.2 Dynamic Multi-view Generation Model

In multi-view learning, data can be represented from different angles or sources, generating multiple views or representations. Common methods for adding perspectives include subgraph sampling, node feature perturbation, edge perturbation, graph structure sampling, and graph structure recombination. Considering the importance of the structural and spatial characteristics of graph data, we adopt subgraph sampling and utilize graph diffusion to generate new views for multi-view learning. By generating new views through graph diffusion, we can better capture the structural and spatial information of graph data, providing a global perspective of the graph structure, thereby improving the model's performance and generalization ability.

The adjacency matrix can be transformed into a diffusion matrix through a normalization operation. Common normalization methods include symmetric normalization and random walk normalization. Symmetric normalization divides each row of the adjacency matrix by the square root of the degree of that row (degree refers to the number of edges connected to the node). Random walk normalization divides each row of the adjacency matrix by the degree of that row. Both normalization methods can convert the adjacency matrix into a diffusion matrix, ensuring that the sum of elements in each row equals 1, thus ensuring the validity of the transition

probabilities. The choice between these two normalization methods depends on the specific application scenario and requirements. Common models include PageRank (PR), Personalized PageRank (PPR) [14], and Heat-kernel PageRank (HK-PR) [34].

To address the limitations of traditional static diffusion matrices, we design a dual-channel dynamic diffusion engine, which serves as a dynamic multi-view generation model.

Local Semantic Preservation Channel: This channel integrates the Heat-kernel diffusion kernel with the Graph Attention Mechanism (GAT) to generate fine-grained local representations. As shown in Equation 1.

$$M_{local} = GAT \odot \sum_{k=0}^{\infty} \frac{h^k e^{-h}}{k!} M^k \frac{\vec{e}}{N} \quad (1)$$

Where h is adaptively adjusted through subgraph density, and HK-PR assigns larger weights to nodes closer to the seed node in terms of distance. $M^k = D^{-1}A$ represents the transition matrix, where D is the degree matrix of the network, and A is the adjacency matrix of the network.

Global Topology Propagation Channel: An improved variant of Personalized PageRank, PPR++, is constructed by introducing node centrality weights, as detailed in Equation 2.

$$M_{global} = \vec{\pi}_s^{PPR++} = (1 - \alpha) \sum_{k=0}^{\infty} \alpha^k (D^{-\gamma} A) \vec{e}_s \quad (2)$$

Where $\gamma = 0.5 + \text{Sigmoid}(\text{MLP}(c_v))$, CV represents the node centrality metric, and $\alpha \in (0, 1)$ is the damping factor, adaptively adjusted based on subgraph density.

The key to multi-view learning lies in integrating information from multiple perspectives, which helps the model capture the underlying structure and features of the data more effectively, thereby improving its robustness and generalization ability. Of course, the design of the model structure is also crucial. To this end, we have designed a momentum-based asynchronous gradient coordination mechanism to extract diverse feature information.

3.3 Momentum Asynchronous Gradient Coordination Mechanism

In graph networks, "asynchronous" typically refers to the different methods of information propagation and neural network weight updates. Different nodes or edges asynchronously and non-blockingly propagate information and update features during the learning process. Traditional graph neural network models usually employ synchronous updates, where all nodes or edges update synchronously at each time step, maintaining consistency in the feature extraction modules. However, for large-scale graphs or graphs with dynamic properties, synchronous updates may lead to decreased computational efficiency and performance bottlenecks. The asynchronous structure decouples the update process of different nodes or edges, allowing them to propagate information and update features at their own pace, thus improving the training efficiency and performance of the graph network. "Asymmetry" refers to the use of different feature extraction methods in the two branches of the neural network, aiming to obtain more comprehensive and diverse feature information, thereby achieving feature decoupling.

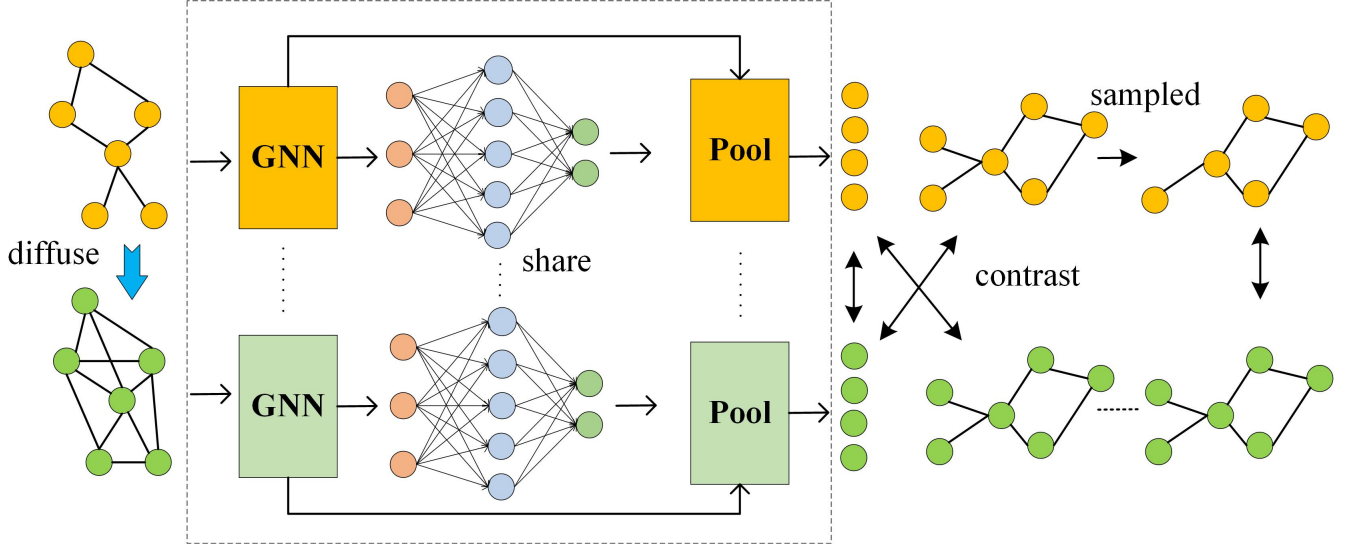


Figure 1: MHGCL Structure Diagram.

In this study, each node or edge in the asynchronous structure propagates information and updates features based on predefined diversified feature extraction modules to acquire more comprehensive feature information. Additionally, weight updates involve gradient updates and momentum updates from different branches. Heterogeneous graph convolution methods such as GraphConv, TAGConv [12], and SAGEConv are used for feature extraction within the asymmetric structure. Therefore, this study proposes a momentum-based asynchronous gradient coordination mechanism. To address the instability problem of training in traditional asynchronous structures, a differentiated gradient propagation strategy is proposed. The main network employs real-time gradient updates using graph convolution methods to capture local temporal features, while the auxiliary network introduces momentum updates ($\beta = 0.95$) and uses graph convolution methods to extract global topological patterns.

The parameter update formula for the main network is shown in Equation 3:

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} L \quad (3)$$

Here, θ represents the parameters of the main network, η is the learning rate, and $\nabla_{\theta} L$ is the gradient of the loss function with respect to the parameters.

The parameter update formula for the auxiliary network is shown in Equation 4:

$$\varphi_t = \beta \varphi_{t-1} - (1 - \beta) \theta_t \quad (4)$$

Here, φ represents the parameters of the auxiliary network, β is the momentum coefficient, which controls the smoothness of the parameter updates.

3.4 Multi-level Contrastive Learning Framework

Existing contrastive learning methods primarily focus on node-to-graph or graph-to-graph comparisons, lacking joint optimization

of multi-level semantics (such as node-subgraph-global graph). To address this, we construct a three-level contrastive loss, which includes cross-view node and graph comparison, node-to-node comparison, and subgraph-to-global graph comparison.

Cross-view node and graph comparison: In View 1, node and graph representations are generated through standard graph convolution (GCN), while in View 2, graph diffusion methods (e.g., PPR) are employed to generate enhanced global-aware node and graph representations. The goal is to improve the node representation's awareness of the global structure. The formula is shown in Equation 5.

$$L_{cross-view} = \max \frac{1}{|G|} \sum_{g \in G} \left[\frac{1}{g} \sum \left[MI(\vec{n}_{local}, \vec{g}_{global}) + MI(\vec{n}_{global}, \vec{g}_{local}) \right] \right] \quad (5)$$

In this context, MI refers to maximizing mutual information, and we adopt Jensen-Shannon Divergence (JSD) as the method for estimating mutual information.

Node-to-node comparison: To enhance the consistency of representations for similar nodes, we perform node-level contrast, as shown in Equation 6.

$$L_{node-node} = \max \frac{1}{|G|} \sum_{g \in G} \left[\frac{1}{g} \sum \left[MI(\vec{n}_{local}, \vec{n}_{global}) \right] \right] \quad (6)$$

Subgraph-to-global graph comparison: To establish the relationship between local structure and global semantics, we design a subgraph-to-global graph comparison. The subgraph is sampled by weighting key subgraphs based on PageRank scores. The formula is shown in Equation 7.

$$L_{sub-global} = \max \frac{1}{|G|} \sum_{g \in G} \left[\frac{1}{g} \sum \left[MI(\vec{g}_{sub}, \vec{g}_{global}) \right] \right] \quad (7)$$

Finally, our three-level contrastive loss is dynamically adjusted by the meta-network. The first 50% of the training focuses on node-level comparisons, while the latter part emphasizes graph-level comparisons. This is illustrated in Equation 8.

$$L_{total} = \lambda_1 L_{cross-view} + \lambda_2 L_{node-node} + \lambda_3 L_{sub-global} \quad (8)$$

The initial value is set to $\lambda_1 = 1.0, \lambda_2 = 0.7, \lambda_3 = 0.5$. Through multi-level contrastive learning, semantic information from nodes, subgraphs, and whole graphs is systematically integrated, significantly enhancing the model's representational capability while ensuring.

Kullback-Leibler Divergence(KLD), also known as relative entropy or information divergence, is a measurement method used to quantify the difference between two probability distributions. KLD can be employed to compare the similarity of two probability distributions or serve as a loss function or regularization term in machine learning. The formula for calculating KLD is shown in Equation 9.

$$KL(P \parallel Q) = \sum P(x) \log \frac{P(x)}{Q(x)} \quad (9)$$

Here, P and Q represent the two probability distributions, while x represents the values of the random variable. A smaller KLD value indicates a closer similarity between the two probability distributions.

Jensen-Shannon Divergence (JSD) is a distance measure used to quantify the similarity between two probability distributions. It is an improved method based on KLD. JSD can be used to compare the similarity of two probability distributions or serve as a loss function or regularization term. Unlike KLD, JSD is symmetric and bounded. The formula for calculating JSD is shown in Equation 10.

$$JSD(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M) \quad (10)$$

Here, P and Q are two probability distributions, M is the average distribution of P and Q , and DKL represents KLD. A smaller JSD value indicates a greater similarity between the two probability distributions.

In the following sections, this paper will explore the effects of different divergences in classification. The different mutual information (MI) estimators are the JSD estimator, KLD estimator, and GAN estimator.

By introducing Alignment and Uniformity [46] to address tolerance and uniformity conflicts. Alignment loss is directly defined as the expected distance between positive sample pairs, while uniformity loss is defined as the average logarithm of paired Gaussian potentials, where t is a fixed parameter. The formulas are shown in Equations 11 and 12.

$$L_{al}(f; \alpha) = E_{(x,y) \sim p_{pos}} [\|f(x) - f(y)\|_2^\alpha], \alpha > 0 \quad (11)$$

$$L_{un}(f; t) = \log E_{x,y \stackrel{i.i.d}{\sim} p_{data}} [e^{-t\|f(x)-f(y)\|_2^2}], t > 0 \quad (12)$$

Alignment is used to measure the proximity between positive sample pairs, encouraging similar features for identical samples. Uniformity assesses the uniformity of the feature distribution on a

hypersphere after regularization, aiming to preserve as much information in the feature distribution as possible. Optimizing these two metrics during training leads to learned features that perform better on downstream tasks, making the loss function more interpretable.

Meanwhile, the temperature coefficient plays a significant role in controlling the penalty strength for hard negative samples. Specifically, a contrastive loss with a smaller temperature tends to penalize the most challenging negative samples more, promoting greater separation of the local structure for each sample and achieving a more even embedding distribution. On the other hand, a contrastive loss with a larger temperature is less sensitive to hard negative samples, and as the temperature approaches positive infinity, the ability to perceive difficult samples gradually diminishes. The paper [45] has already demonstrated that a reasonable temperature coefficient can address the trade-off dilemma between tolerance and uniformity.

4 Experiments

In this section, we will validate the exceptional performance of our model in node classification and graph classification tasks using different types of data, thereby highlighting the model's versatility across various downstream tasks. Improving feature diversity through an asymmetric structure and a loss function that integrates global and local features; addressing the Uniformity-Tolerance Dilemma by introducing a temperature coefficient, Alignment, and Uniformity. We not only compare our model extensively with state-of-the-art models but also conduct ablation experiments from various perspectives to analyze the roles played by different components in the model.

4.1 Dataset

To validate the classification accuracy of the model for node classification and graph classification, we used seven datasets for verification: citation datasets Cora [33], Citeseer [16], Pubmed [39], bioinformatics datasets MUTAG[10], D&D [11], PTC-MR [29], and social network dataset IMDB-B [52]. These datasets serve as multiple downstream applications to test the model's broad application capabilities.

4.2 Experimental Setup

Our experimental design is as follows: Each experiment is performed in a loop for ten times, and the best result is selected. We use a linear model for evaluation and record the average classification accuracy and standard deviation of the tests. For the node classification task, we choose 300 epochs, and for the graph classification task, we choose 30 epochs, with an early termination mechanism set up simultaneously. The batch size is set to 64, the hidden layer is designed with 512 nodes, the momentum is set to 0.9 and the learning rate is set to 0.001. With this experimental design, we can assess the performance of the model on different tasks and select the best results.

4.3 Evaluation Metrics

Standard deviation is a statistical measure that quantifies the dispersion or variability of data, playing a crucial role in model evaluation. When the standard deviation is small, it indicates that the data

has less fluctuation, and the model exhibits relatively stable performance and good generalization ability across different data samples. If two models have similar average accuracy but one model has a smaller standard deviation, we can consider that the model with a smaller standard deviation is relatively more reliable and stable. To this end, this paper primarily utilizes two evaluation metrics, namely standard deviation and accuracy, to assess the performance of the model. The formula for the standard deviation is shown in Equation 13.

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} \quad (13)$$

Here, x represents the i th data point, μ represents the mean of the data points and $\sum$ represents summation.

4.4 Comparison with State-of-the-Art Results

In this experiment, we use a linear classifier to validate the classification results. Firstly, each model undergoes ten iterations and saves the best model parameters. Then, during the validation phase, we perform another ten iterations and take the average value as the final result. Finally, we calculate the standard deviation of the validation results to assess the stability and consistency of the outcomes. This experimental design helps us obtain reliable results during model training and validation, allowing for an accurate assessment of model performance.

We conducted model validation in node classification tasks using three different citation datasets. To evaluate the advancements of our model, we performed comparative experiments with advanced models.

Table 1: Accuracy and Standard Deviation in Node Classification Tasks.

	Model	Cora	Citeseer	Pubmed
unsupervised	MVGRL [19]	83.2±2.4	73.3±2.3	80.1±2.0
	DGI [44]	81.6±3.0	71.8±3.0	76.1±1.0
	GAE [28]	71.5±0.4	65.8±0.4	72.1±0.5
	InfoGCL [51]	83.5±0.3	73.5±0.4	79.1±0.2
	GREET [32]	83.8±0.9	73.1±0.8	80.3±1.0
	MHGCL	84.5±0.1	74.5±0.1	81.6±0.1
supervised	GAT [43]	83.0±2.5	72.5±2.6	79.0±1.0
	GCN [27]	81.4±3.2	70.3±2.5	79.0±1.0
	GraphSAGE [18]	81.8±2.7	71.5±2.8	78.7±3.1
	APPNP [15]	83.2±2.6	72.6±3.1	79.5±1.0

From the table 1, it can be observed that although MHGCL is a self-supervised model, it has achieved comparable results to supervised models, which is an exciting discovery for self-supervised research. Compared to MVGRL, our model demonstrates significant improvements across all three datasets, with increases of 1.3%, 1.2%, and 1.5%, respectively. It also outperforms other models by a considerable margin. Importantly, MHGCL exhibits significantly lower standard deviation than other models, indicating its strong stability and robustness.

In graph classification tasks, we used four different datasets to validate our model. To assess the advancements of our model, we selected several state-of-the-art models as comparison methods.

Table 2: Accuracy and Standard Deviation in Graph Classification Tasks.

Model	MUTAG	PTC-MR	IMDB-B	D&D
MVGRL [19]	89.7±5.6	62.2±6.3	70.4±3.4	79.3±2.3
Sub2Vec [1]	61.1±15.8	60.0±6.4	55.3±1.5	54.3±2.4
Graph2Vec [35]	83.2±9.6	60.2±6.9	71.1±0.5	70.3±2.3
MSSGCL [31]	89.7±0.6	-	73.1±0.4	79.7±0.4
SimGRACE [50]	89.0±1.3	-	73.1±0.4	77.4±1.1
InfoGraph [41]	89.0±1.1	61.7±5.4	73.0±0.9	72.9±1.8
SEGA [48]	90.2±0.7	62.6±5.1	73.6±0.4	78.9±0.6
MHGCL	91.5±3.6	63.3±5.2	74.3±2.6	80.2±1.9

From the table 2, compared to MVGRL, our model achieved improvements of 1.2%, 1.1%, and 0.9% in accuracy on the MUTAG, PTC-MR, and D&D datasets, respectively. Additionally, our model outperformed InfoGraph by 1% in accuracy on the IMDB-B dataset. Furthermore, our model exhibited a relatively smaller standard deviation compared to other models.

Despite multiple experiments, PTC-MR still presents large result variance. The root cause lies in the dataset's inherent molecular structural complexity and diversity, which trigger classification accuracy fluctuations and thus lead to high variance.

Overall, MHGCL has achieved good performance in two different classification tasks. To validate the model's performance on datasets of varying magnitudes, we intentionally selected datasets of different magnitudes to assess the robustness and applicability of the model.

4.5 Ablation Study

In order to gain a deeper understanding of the working principles of the model and the contributions of its key components to overall performance, as well as to evaluate their roles, we designed three different ablation experiments: asynchronous asymmetry, divergence calculation methods, and contrastive loss calculation methods. Through these experiments, we are able to enhance our understanding of the model, optimize its performance, and drive advancements in the related research field.

Asymmetric structures, through convolution in different layers, can extract richer and more diverse feature information. Compared to symmetric structures, asymmetric structures allow the network to better capture local, global, and multi-scale features. To further explore their outstanding performance, we conducted multiple groups of ablative experiments on asynchronous asymmetric network structures. We designed five control groups, including GraphConv (G), TAGConv (T), and SAGEConv (S). The experimental results show that, overall, the accuracy of asymmetric structures (G+T, G+S, T+S) is slightly higher than that of symmetric structures (G+G, T+T, S+S), and they exhibit greater stability in terms of standard deviation. These experiments confirm that asymmetric structures can extract richer feature information and improve model accuracy. By performing convolution at different levels, asymmetric

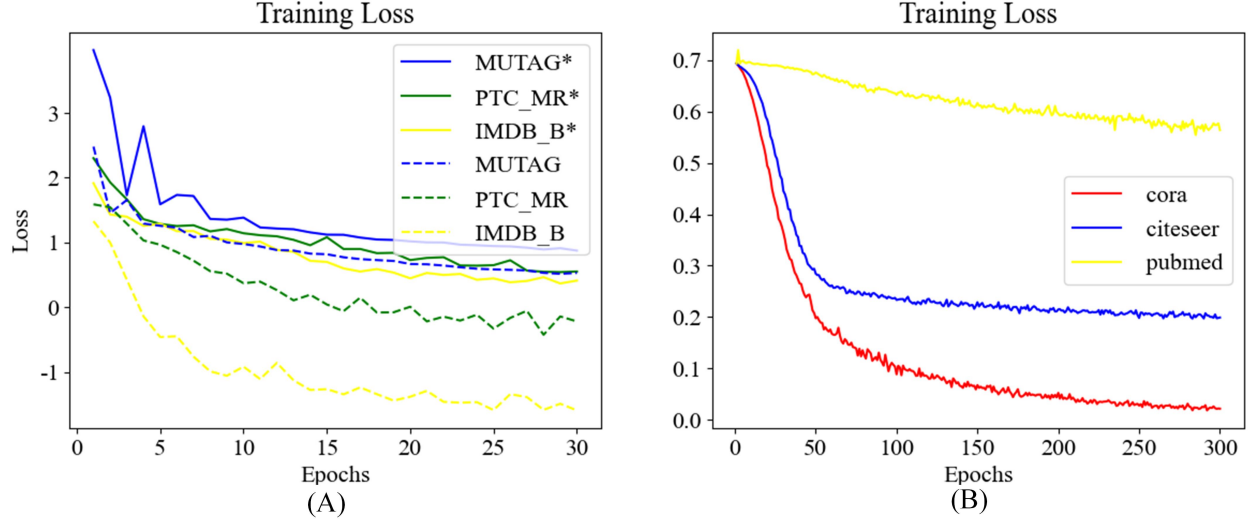


Figure 2: Graph Classification Loss Variation Curve and Nodes Classification Loss Variation Curve.

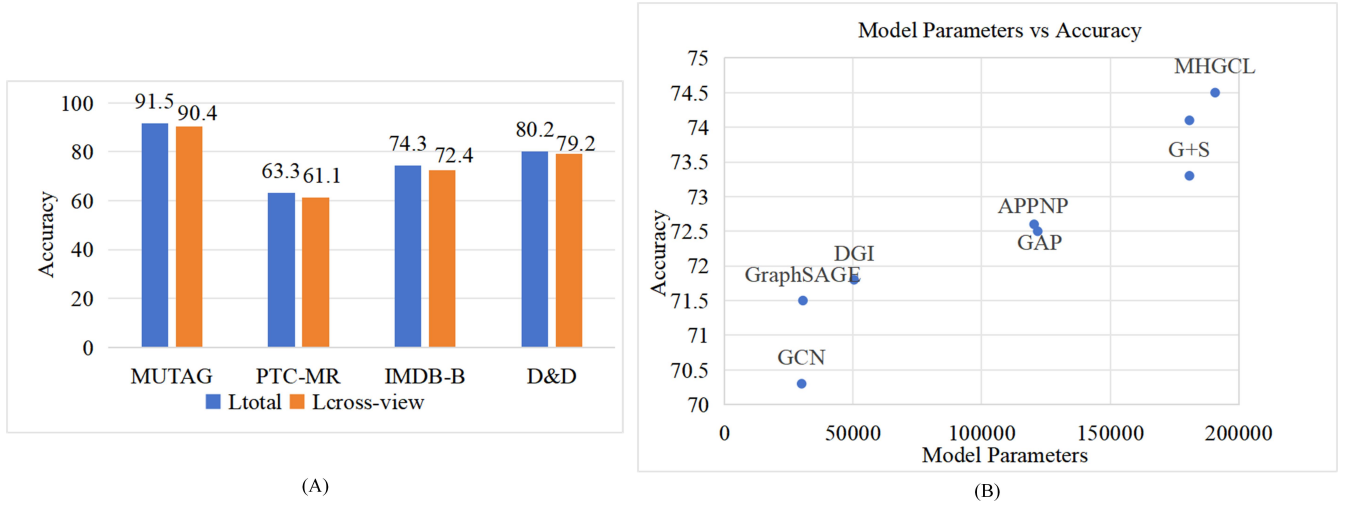


Figure 3: (A) The analysis of the loss function results (B) Model Parameters Vs Accuracy.

structures enable the network to better capture local, global, and multi-scale features. This means that asymmetric structures can flexibly capture various features in graph data, thus providing a more comprehensive and rich feature representation.

From the table 3, comparing different divergence calculation methods, it was found that JSD (Jensen-Shannon Divergence) converges faster and achieves a stable and smaller loss value compared to other divergence calculation methods. The GG and GT ablation experiments also demonstrated the superiority of asymmetric structures, indicating that adopting asymmetric structures in graph representation learning can lead to better performance. Lttotal represents the proposed three-level hierarchical contrastive learning

Table 3: Ablation experiments in Node Classification Tasks.

Structures	Cora	Citeseer	Pubmed
G+T	84.5±0.1	74.5±0.1	81.6±0.1
G+S	83.4±0.1	74.1±0.1	77.6±0.2
T+S	84.0±0.1	74.0±0.1	80.0±0.1
G+G	83.2±2.4	73.3±2.3	80.1±2.0
T+T	83.1±0.1	73.7±0.1	80.9±0.1
S+S	77.0±0.1	72.0±0.1	77.6±0.1

framework, while Lcross-view corresponds specifically to the cross-view node and graph comparison component. Lttotal effectively

integrates local and global information through joint optimization at node, subgraph, and global graph levels, providing comprehensive data characterization while capturing essential feature representations. This hierarchical architecture enhances the model's expressive power and discriminative capability, ultimately achieving superior classification accuracy. The results further validate the strong superiority of the proposed contrastive loss calculation method presented in this work.

Table 4: Ablation experiments in Graph Classification Tasks.

	MUTAG	PTC-MR	IMDB-B	D&D
JSD	91.5±3.6	63.3±5.2	74.3±2.6	80.2±1.9
GAN	90.4±7.8	59.5±8.2	73.5±3.7	79.6±2.3
KL	89.9±9.8	58.7±4.2	73.2±4.1	79.0±3.1
GT	91.5±3.6	63.3±5.2	74.3±2.6	80.2±1.9
GG	88.3±9.1	57.5±6.5	72.5±3.5	79.1±2.2
$L-\beta$	90.6±7.2	62.2±8.6	72.6±4.7	79.8±5.8
Lcross-view	90.4±6.1	61.1±8.5	72.4±3.7	79.2±2.8
Ltotal	91.5±3.6	63.3±5.2	74.3±2.6	80.2±1.9

From the table 4, the model's accuracy improved while the variance decreased. The experimental results demonstrate that adopting an asynchronous asymmetric architecture, coupled with appropriate divergence computation methods and multi-level contrastive loss calculation strategies, achieves superior performance and accuracy in graph representation learning. The applicability and effectiveness of these approaches further validate their significance in handling complex graph-structured data, providing robust theoretical and empirical support for advancing research and practical applications of graph neural networks. When the momentum parameter β is removed, the variance fluctuation of the model increases significantly. This phenomenon directly demonstrates that the momentum parameter plays a role in smoothing parameter updates and enhancing model stability.

4.6 Discussion

The results of this study have provided new insights and perspectives for understanding key issues in a specific field. In this discussion section, we will delve into these results in detail and place them within a broader context to emphasize their significance and potential applications.

Because the loss value is the positive expectation minus the negative expectation, there can be situations where the loss value is negative. To optimize the loss value, adding a temperature coefficient as well as Alignment and Uniformity can improve the value of the loss, making it more stable. This is validated by the loss curve, where the curve is smoother and converges better. In Figure 2(A), "MUTAG*, IMDB*, PTC-MR*" has been optimized, while "MUTAG, IMDB, PTC-MR" represents the scenario without the inclusion of temperature coefficients, alignment and uniformity.

In Figure 2(B), it can be observed that the overall loss values in the experiment are relatively small, indicating strong performance on the training data and a successful fit to the data. The loss curve exhibits a smooth descent and converges within a certain range,

signifying that the model fits well, and its predictive capabilities continue to improve throughout the training process.

MHGCL also faces a common issue in contrastive learning, namely the large number of model parameters. It is still a worthwhile research problem to effectively extract features, contrast positive and negative samples, while reducing the model parameter size. To address this, we compared the relationship between model size and accuracy under the same input parameters, as shown in Figure 3(A). As the contrastive model itself operates on two separate pathways, the increase in parameter count is entirely acceptable. The main reason for the disparity lies in the inherent differences in graph convolutional networks themselves, which are independent of the framework. As illustrated in Figure 3(B), the model size is presented.

5 Conclusions

In this paper, we proposed MHGCL, a multi-view hierarchical graph contrastive learning framework built upon an asynchronous asymmetric architecture. By jointly addressing insufficient feature diversity, incomplete semantic hierarchy modeling, and the tolerance-consistency trade-off, MHGCL provides a unified solution for self-supervised graph representation learning. Specifically, we introduced a dynamic multi-view generation model based on a dual-channel diffusion engine that integrates local semantic preservation via heat-kernel diffusion and graph attention with global topology propagation through a centrality-weighted Personalized PageRank variant. This design enables the generation of diverse yet semantically consistent graph views, overcoming the limitations of static diffusion-based augmentations. To further decouple local and global semantic learning, we proposed an asynchronous asymmetric learning architecture that employs heterogeneous graph convolutional networks with differentiated gradient propagation strategies. A momentum-based gradient coordination mechanism was introduced to stabilize training, effectively mitigating representation collapse and improving robustness. Moreover, we designed a hierarchical contrastive learning framework that jointly optimizes node-level, subgraph-level, and graph-level semantics through a three-tier contrastive loss. By dynamically adjusting loss weights and incorporating alignment, uniformity, and temperature-aware regulation, the framework achieves a better balance between local tolerance and global consistency. Extensive experiments on node and graph classification benchmarks validate the effectiveness and generalization capability of MHGCL across datasets of varying scales and domains.

Although MHGCL shows stable improvements across benchmarks, several directions remain. Future work may explore lightweight implementations to reduce computational overhead, extend the framework to dynamic or temporal graphs via time-aware diffusion or contrastive mechanisms, and investigate cross-domain transfer learning through large-scale pretraining and downstream adaptation to further enhance generalization and practical applicability.

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